Mahalingam Ramasamy

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Technical Skills

Languages: Java, TypeScript, JavaScript, Python, SQL
Backend: Node.js, NestJS, Spring Boot, FastAPI, REST APIs, WebSockets, SSE, Celery
Databases: PostgreSQL, pgvector, Redis, Prisma
Cloud: AWS (ECS Fargate, ALB, Lambda, S3, CloudFront, SES, VPC, Secrets Manager), GCP
DevOps: Terraform, Docker, GitHub Actions, CI/CD
Frontend: React, Next.js, Tailwind CSS

Experience

Ekko Medical Solutions Pvt Ltd **Oct 2025 – Present**
Software Developer (Backend) *Chennai, India*

- Cut AI clinical-note generation time by **50% (from 50s to 25s)** by routing transcript refinement to a lightweight LLM and generating secondary note sections asynchronously via an event-driven pipeline (NestJS).
- Built a horizontally scalable real-time transcription microservice (**WebSockets, Redis, PostgreSQL**) delivering sub-second live transcripts for sessions up to **5 hours**, with automatic recovery on disconnect.
- Designed a provider-agnostic LLM layer across **OpenAI, Gemini and Anthropic** powering **15+ AI tasks**, routing each task to the best-fit model with same-request failover to a backup provider.
- Built per-clinic medicine matching that corrects misheard drug names in AI prescriptions using **PostgreSQL fuzzy and phonetic search** (Levenshtein, Double Metaphone), with dose-safety rules and **zero extra LLM calls**.
- Built **Cashfree** subscription billing with **UPI/eNACH autopay across 7 plans**, HMAC-verified webhooks, idempotent processing and hourly reconciliation.
- Migrated production from AWS us-east-1 to **ap-south-1 (Mumbai)** across 3 environments with **Terraform** and **ECS Fargate** in a 1-hour window, with GitHub Actions CI/CD.

ROX Hi-TECH **June 2024 – Oct 2025**
Full Stack Developer *Chennai, India*

- Built an HR workflow platform in **Java/Spring Boot** serving **100,000+ users** for leave, transfer and reviews.
- Designed a configurable approval engine across **6+ request types**, launching new workflows without code changes.
- Built **Spring Boot REST APIs** integrating **SAP ERP** via **OData** to sync HR master data.
- Implemented **RBAC** across **3 user roles** with **Spring Security**, working in **Agile** sprints with **JUnit** tests.

Projects

Shoppy – E-Commerce Platform | *Java, Spring Boot, Next.js, PostgreSQL, Stripe, AWS* **Live** | **GitHub**

- Wrote **54 automated tests** (JUnit 5, Mockito, Testcontainers) at **80% coverage**, run in CI on every push.
- Integrated **Stripe** with signature-verified, **idempotent webhooks** for exactly-once order creation.
- Secured APIs with **Spring Security, JWT** and RBAC; pushed live product updates via **WebSockets (STOMP)**.
- Automated **AWS** deploys via CodePipeline and **Elastic Beanstalk**, with RDS PostgreSQL, S3 and CloudFront.

MedRAG – Medical RAG Assistant | *Python, FastAPI, React, pgvector, Gemini, RAGAS* **Live** | **GitHub**

- Built **hybrid search** (pgvector and Postgres full-text) fused via **Reciprocal Rank Fusion**.
- Re-ranked **top 20** to **top 5** retrieval candidates with a **cross-encoder** (ms-marco-MiniLM).
- Achieved **0.93 context recall** and **0.86 faithfulness** on a **RAGAS** evaluation.
- Streamed grounded, cited **Gemini** answers over **SSE** on a **512 MB** instance using **ONNX** embeddings.

Education

SASTRA Deemed University **2020 – 2024**
B.Tech, Computer Science and Business Systems *Thanjavur, India*